Linear Algebra — Lecture 3

Linear Transformation, Null Space and Range, the Matrix Representation

Friedberg–Insel–Spence, *Linear Algebra* 4e, §2.1, §2.2
D. C. Lay, *Linear Algebra and Its Applications* 4e, §1.8, §1.9, §4.2, §4.4, §5.4

Topics. Linear transformation, null space and range, the matrix representation of a linear transformation.

How the two texts divide the work. FIS (the main text) develops the definitions and the theorems for **transformations between arbitrary vector spaces**, and it alone carries the whole theory of nullity, rank, the dimension theorem and the injectivity test; Lay stays in $\mathbb{R}^n \to \mathbb{R}^m$ with matrix transformations and supplies the geometric intuition and the standard matrix, and does *not* cover the theory of the null space and the range. Mapping for this lecture:

Topic	Lay 4e (printed)	Friedberg 4e (printed)
Linear transformation	§1.8 (p. 62–69)	§2.1 (p. 65)
Geometric examples	§1.8 (p. 65)	§2.1 (p. 66)
Null space and range	§4.2 (p. 198)	§2.1 (p. 67)
Nullity and rank	—	§2.1 (p. 69–70)
Testing injectivity	§1.8 exercises (p. 68–69)	§2.1 (p. 71)
Determined by a basis	—	§2.1 (p. 72)
Matrix representation	§1.9 (p. 70), §5.4 (p. 288)	§2.2 (p. 80)

Two questions that drive the whole lecture: (1) *Linearity*: is this map linear? The test is a **single** equation, not a check of the two axioms one by one. (2) *Null space and range*: which directions get crushed to zero, and which part gets filled up? Those two decide which columns of the matrix are redundant.

How to read: use Lay's geometric examples and the standard matrix to build the picture (computational layer), then Friedberg's definitions and theorems to see why each step is legitimate (proof layer).

All page numbers here are printed pages. For Lay, PDF page = printed page +17; for Friedberg, PDF page = printed page +15.

1 Linear Transformations

Definition 1.1 (Linear transformation; Friedberg, §2.1, p. 65)**.** Let V and W be vector spaces (over $\mathbb{F}$). A function $T : V \to W$ is called a **linear transformation** from V to W if for all $x, y \in V$ and $c \in \mathbb{F}$,

(a) $T(x + y) = T(x) + T(y)$ (it preserves addition);
(b) $T(cx) = cT(x)$ (it preserves scalar multiplication).

Here V is the *domain* of T and W its *codomain*; $T(x)$ is the **image** of x, and the set of all images is the **range** of T. Lay, §1.8, p. 63.

Proposition 1.1 (Four properties to keep at hand; Friedberg lists them in §2.1, p. 65, Exercise 7 on p. 75)**.** *Let $T : V \to W$ be a function. Then*

(1) if T is linear then $T(\mathbf{0}_V) = \mathbf{0}_W$;
(2) T is linear if and only if $T(cx + y) = cT(x) + T(y)$ for all $x, y \in V$ and $c \in \mathbb{F}$;
(3) if T is linear then $T(x - y) = T(x) - T(y)$;

(4) for every positive integer n and all $x_1, \dots, x_n \in V$, $a_1, \dots, a_n \in \mathbb{F}$: T is linear $\iff$ $T\left(\sum_{i=1}^{n} a_i x_i\right) = \sum_{i=1}^{n} a_i T(x_i)$.

Proof. **(1)** By definition 1.1(a), $T(\mathbf{0}_V) = T(\mathbf{0}_V + \mathbf{0}_V) = T(\mathbf{0}_V) + T(\mathbf{0}_V)$. Adding $-T(\mathbf{0}_V)$ to both sides gives $T(\mathbf{0}_V) = \mathbf{0}_W$.

(2) ($\Leftarrow$) Suppose that one equation holds for all x, y, c. Take first $c = 1$, $y = \mathbf{0}$:

$$T(\mathbf{0}_V) = T(1\mathbf{0}_V + \mathbf{0}_V) = 1 \cdot T(\mathbf{0}_V) + T(\mathbf{0}_V),$$

so $T(\mathbf{0}_V) = \mathbf{0}_W$. Now take $y = \mathbf{0}$: $T(cx) = T(cx + \mathbf{0}_V) = cT(x) + T(\mathbf{0}_V) = cT(x)$, which is (b); then take $c = 1$: $T(x + y) = T(1x + y) = T(x) + T(y)$, which is (a).

(2) ($\Rightarrow$) By (a) and (b), $T(cx + y) = T(cx) + T(y) = cT(x) + T(y)$.

(3) By (a) and (b),

$$T(x - y) = T\big(x + (-1)y\big) = T(x) + (-1)T(y) = T(x) - T(y).$$

(4) ($\Rightarrow$) Induct on n. The case $n = 1$ is (b): $T(a_1 x_1) = a_1 T(x_1)$. If it holds for $n - 1$, split off the last term:

$$T\left(\sum_{i=1}^{n} a_i x_i\right) = T\left(\left(\sum_{i=1}^{n-1} a_i x_i\right) + a_n x_n\right) = T\left(\sum_{i=1}^{n-1} a_i x_i\right) + T(a_n x_n) = \sum_{i=1}^{n-1} a_i T(x_i) + a_n T(x_n),$$

where the second equality is (a) and the third uses the induction hypothesis on the first part and (b) on $T(a_n x_n)$.

(4) ($\Leftarrow$) Take $n = 1$ (with $x_1 = x$, $a_1 = c$) to get (b), and $n = 2$ with $a_1 = a_2 = 1$ to get (a). Hence T is linear. □

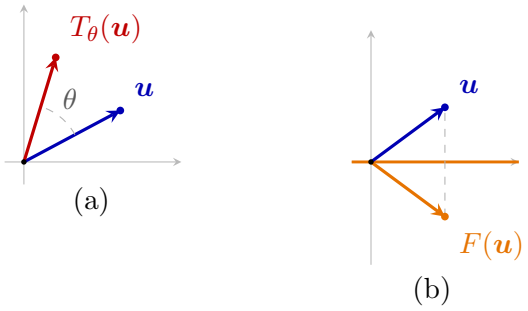
Remark 1.1 (Additivity alone is not enough; Friedberg §2.1, p. 65 and Exercises 37–39, p. 78)**.** Between vector spaces over $\mathbb{Q}$, every **additive** function ($T(x + y) = T(x) + T(y)$) is automatically linear (Friedberg Exercise 37, p. 78). But once $\mathbb{R}$ is regarded as a vector space over $\mathbb{Q}$ there are uncountably many additive functions and *most of them are not linear*: Exercise 38–39 (p. 78) take complex conjugation, and swap two vectors of a basis.

Approach. To test linearity, always use the single equation of proposition 1.1(2): substitute $cx+y$ on one side, evaluate $cT(x) + T(y)$ on the other, and compare. Do not go back to (a) and (b) separately: together they are only two special cases of that one equation, whereas the one equation covers *arbitrary* x, y, c at once.

Definition 1.2 (Identity and zero transformations; Friedberg §2.1, p. 67)**.** For vector spaces V, W the **identity transformation** $I_V : V \to V$ is defined by $I_V(x) = x$, and the **zero transformation** $T_{\mathbf{0}} : V \to W$ by $T_{\mathbf{0}}(x) = \mathbf{0}$. Both are linear.

Remark 1.2 (Geometric picture: rotation, reflection, projection; Friedberg Figure 2.1, p. 66; Lay §1.9, Tables 1–4, p. 73–75)**.** The three most important geometric linear transformations of $\mathbb{R}^2$ are the counterclockwise rotation by θ, the reflection about the x-axis, and the projection onto the x-axis:

$$T_\theta \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} a_1 \cos\theta - a_2 \sin\theta \\ a_1 \sin\theta + a_2 \cos\theta \end{bmatrix}, \qquad F \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} a_1 \\ -a_2 \end{bmatrix}, \qquad P \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} a_1 \\ 0 \end{bmatrix}.$$



(a) A rotation: length is preserved and the direction turns through θ; (b) A reflection: folding about the x-axis, every point of that axis is fixed.

Remark 1.3 (The usual linear transformations and the usual non-linear tells; Friedberg Examples 5–7, p. 67 and Exercise 9, p. 75; Lay §1.8 Exercises 23–24, p. 69)**.**

- **Linear at the level of the formula**: the transpose $T(A) = A^\top$ (Friedberg Example 5, p. 67); differentiation $T(f) = f'$ (Example 6, p. 67); the definite integral $T(f) = \int_a^b f(t)\,dt$ (Example 7, p. 67). Linearity follows directly from the fact that differentiation and integration are taken on linear combinations.
- **Linear at the geometric level**: rotation, reflection, projection and shear (Lay §1.8 Examples 2–3, p. 65), whereas a **translation** is not: $T(x) = Ax + b$ fails to be linear when $b \neq \mathbf{0}$ (Lay Exercises 23–24, p. 69).
- **These are the danger signals**: a constant term, $|x_i|$, $\sin x_i$, a product of two variables, x_i^2. Friedberg Exercise 9 (p. 75) and Lay Exercises 32–33 (p. 69) are all of this kind; the test is that $T(\mathbf{0}) \neq \mathbf{0}$ or $T(cx + y) \neq cT(x) + T(y)$ fails somewhere.

2 Null Space and Range

Definition 2.1 (Null space (kernel) and range; Friedberg, §2.1, p. 67)**.** Let V, W be vector spaces and $T : V \to W$ linear. The **null space** (or **kernel**) of T is the set of those vectors of V that are mapped to the zero vector, written $\mathcal{N}(T)$:

$$\mathcal{N}(T) = \{x \in V : T(x) = \mathbf{0}\}.$$

The **range** (or **image**) of T is the set of all images of vectors of V, written $\mathcal{R}(T) = \{T(x) : x \in V\}$.

Theorem 2.1 (Null space and range are subspaces; Friedberg Theorem 2.1, §2.1, p. 68)**.** *Let V, W be vector spaces and $T : V \to W$ linear. Then $\mathcal{N}(T)$ is a subspace of V and $\mathcal{R}(T)$ is a subspace of W.*

Proof. To keep the notation clean, write $\mathbf{0}_V$ and $\mathbf{0}_W$ for the zero vectors of the two spaces.

By proposition 1.1(1), $T(\mathbf{0}_V) = \mathbf{0}_W$, so $\mathbf{0}_V \in \mathcal{N}(T)$. If $x, y \in \mathcal{N}(T)$ and $c \in \mathbb{F}$, then $T(x + y) = T(x) + T(y) = \mathbf{0}_W + \mathbf{0}_W = \mathbf{0}_W$ and $T(cx) = cT(x) = c\mathbf{0}_W = \mathbf{0}_W$. Hence $x + y \in \mathcal{N}(T)$ and $cx \in \mathcal{N}(T)$, so $\mathcal{N}(T)$ is a subspace of V.

Again $T(\mathbf{0}_V) = \mathbf{0}_W$, so $\mathbf{0}_W \in \mathcal{R}(T)$. If $x, y \in \mathcal{R}(T)$ and $c \in \mathbb{F}$, there are $v, w \in V$ with $T(v) = x$ and $T(w) = y$, so $T(v + w) = T(v) + T(w) = x + y$ and $T(cv) = cT(v) = cx$. Hence $x + y \in \mathcal{R}(T)$ and $cx \in \mathcal{R}(T)$, so $\mathcal{R}(T)$ is a subspace of W.

The two halves differ only in direction: on the kernel side "x, y lie in the set" can be used directly; on the range side one must first **pull representatives back** to $v, w \in V$ before linearity can be used to move the operations into V. □

Definition 2.2 (Left-multiplication transformation; Friedberg, §2.3, p. 92; Lay, §1.8, p. 63 calls it a matrix transformation)**.** Let A be an $m \times n$ matrix. Define $L_A : \mathbb{R}^n \to \mathbb{R}^m$ by $L_A(x) = Ax$. The map L_A is called the **left-multiplication transformation** of A.

Proposition 2.2 (For a left-multiplication transformation the kernel and the range are the null space and the column space; Lay §4.2, p. 199 and p. 201; Friedberg §2.1, p. 67)**.** *Let A be an $m \times n$ matrix. Then*

$$\mathcal{N}(L_A) = \operatorname{Nul} A, \qquad \mathcal{R}(L_A) = \operatorname{Col} A.$$

Proof. $A\boldsymbol{x} = \mathbf{0}$ says exactly that x is mapped by L_A to the zero vector, so $\mathcal{N}(L_A) = \operatorname{Nul} A$ (Lay §4.2, p. 199).

On the other hand $L_A(x) = Ax = x_1\boldsymbol{a}_1 + \cdots + x_n\boldsymbol{a}_n$, where $\boldsymbol{a}_j$ is the jth column of A; and the column space of A is by definition the set of all linear combinations of its columns (Lay §4.2, p. 201). Hence $\mathcal{R}(L_A) = \operatorname{Col} A$. □

Theorem 2.3 (The range is the span of the images of a basis; Friedberg Theorem 2.2, §2.1, p. 68)**.** *Let V, W be vector spaces and $T : V \to W$ linear. If $\beta = \{v_1, \ldots, v_n\}$ is a basis for V, then*

$$\mathcal{R}(T) = \operatorname{Span}\big(T(\beta)\big) = \operatorname{Span}\{T(v_1), T(v_2), \ldots, T(v_n)\}.$$

Proof. Each $T(v_i)$ is an image of a vector, so each lies in $\mathcal{R}(T)$; and $\mathcal{R}(T)$ is a subspace (theorem 2.1), so it contains every linear combination of the $T(v_i)$. That is $\mathcal{R}(T) \supseteq \operatorname{Span}(T(\beta))$.

Conversely, take any $w \in \mathcal{R}(T)$, so $w = T(v)$ for some $v \in V$. Expand v in the basis β: $v = a_1v_1 + \cdots + a_nv_n$. By linearity,

$$w = T(v) = T\left(\sum_{i=1}^{n} a_iv_i\right) = \sum_{i=1}^{n} a_iT(v_i) \in \operatorname{Span}(T(\beta)),$$

so $\mathcal{R}(T) \subseteq \operatorname{Span}(T(\beta))$.

The two inclusions give the equality. The basis may be replaced by any spanning set: neither direction of the argument uses uniqueness. □

Definition 2.3 (Nullity and rank; Friedberg, §2.1, p. 69)**.** Let $T : V \to W$ be linear. If $\mathcal{N}(T)$ and $\mathcal{R}(T)$ are both finite-dimensional, then $\operatorname{nullity}(T) = \dim \mathcal{N}(T)$ is called the **nullity** of T and $\operatorname{rank}(T) = \dim \mathcal{R}(T)$ its **rank**.

Theorem 2.4 (Dimension Theorem; Friedberg Theorem 2.3, §2.1, p. 70)**.** *Let $T : V \to W$ be linear. If V is finite-dimensional, then*

$$\operatorname{nullity}(T) + \operatorname{rank}(T) = \dim V.$$

Proof. Let $\dim V = n$, $\dim \mathcal{N}(T) = k$, and take a basis $\{v_1, \ldots, v_k\}$ for $\mathcal{N}(T)$. By the Replacement Theorem of the Lecture 2 notes, extend it to a basis $\beta = \{v_1, \ldots, v_n\}$ for V. What has to be shown is that

$$S = \{T(v_{k+1}), \ldots, T(v_n)\}$$

is a basis for $\mathcal{R}(T)$.

Spanning: by theorem 2.3 and $T(v_i) = \mathbf{0}$ for $1 \le i \le k$,

$$\mathcal{R}(T) = \operatorname{Span}\{T(v_1), \ldots, T(v_n)\} = \operatorname{Span}\{T(v_{k+1}), \ldots, T(v_n)\} = \operatorname{Span}(S).$$

Independence: suppose $\sum_{i=k+1}^{n} b_iT(v_i) = \mathbf{0}$. By linearity, $T\left(\sum_{i=k+1}^{n} b_iv_i\right) = \mathbf{0}$, so $\sum_{i=k+1}^{n} b_iv_i \in \mathcal{N}(T)$. There are therefore $c_1, \ldots, c_k \in \mathbb{F}$ with $\sum_{i=k+1}^{n} b_iv_i = \sum_{i=1}^{k} c_iv_i$, that is

$$\sum_{i=1}^{k}(-c_i)v_i + \sum_{i=k+1}^{n} b_iv_i = \mathbf{0}.$$

Since $v_1, \ldots, v_n$ is a basis, all coefficients vanish; in particular $b_{k+1} = \cdots = b_n = 0$. So S is linearly independent.

Now $\mathcal{R}(T) = \mathrm{Span}(S)$ gives $\mathrm{rank}(T) = n - k$, while $k = \mathrm{nullity}(T)$ and $n = \dim V$, which is the formula. □

Remark 2.1 (Null spaces and ranges of the basic examples; Friedberg Examples 8–9, p. 68)**.**

- Identity and zero transformations: $\mathcal{N}(I_V) = \{\mathbf{0}\}$, $\mathcal{R}(I_V) = V$; $\mathcal{N}(T_\mathbf{0}) = V$, $\mathcal{R}(T_\mathbf{0}) = \{\mathbf{0}\}$. This is why the definitions single out these two transformations by name.
- $T : \mathbb{R}^3 \to \mathbb{R}^2$, $T(a_1, a_2, a_3) = (a_1 - a_2, 2a_3)$: from $a_1 = a_2$ and $a_3 = 0$ we get $\mathcal{N}(T) = \{(a, a, 0) : a \in \mathbb{R}\}$; the two components of the image range independently over $\mathbb{R}$, so $\mathcal{R}(T) = \mathbb{R}^2$. Hence $\mathrm{nullity}(T) = 1$ and $\mathrm{rank}(T) = 2$, and $1 + 2 = 3 = \dim \mathbb{R}^3$, so the dimension theorem holds.

This is the step that turns the abstract $\mathcal{N}(T)$ into the homogeneous systems of Lecture 1: when $T(x) = Ax$, the identity $\mathcal{N}(T) = \mathrm{Nul}\, A$ says that $\mathcal{N}(T)$ is the solution set of $A\boldsymbol{x} = \mathbf{0}$, which row reduction produces directly; and $\mathcal{R}(T) = \mathrm{Col}\, A$ says that taking the **pivot columns** of A gives a basis (Lay §4.3, p. 212).

Theorem 2.5 (Testing for injectivity; Friedberg Theorem 2.4, §2.1, p. 71)**.** *Let V, W be vector spaces and $T : V \to W$ linear. Then T is **one-to-one** if and only if $\mathcal{N}(T) = \{\mathbf{0}\}$.*

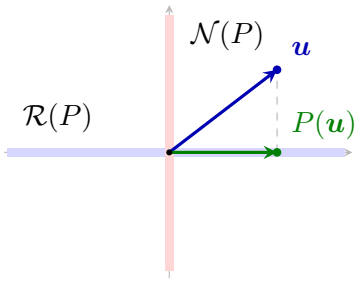
Proof. ($\Rightarrow$) Suppose T is one-to-one and $x \in \mathcal{N}(T)$. Then $T(x) = \mathbf{0} = T(\mathbf{0}_V)$, so injectivity gives $x = \mathbf{0}_V$ and $\mathcal{N}(T) = \{\mathbf{0}\}$.

($\Leftarrow$) Suppose $\mathcal{N}(T) = \{\mathbf{0}\}$. If $T(x) = T(y)$, then by proposition 1.1(3),

$$\mathbf{0} = T(x) - T(y) = T(x - y),$$

so $x - y \in \mathcal{N}(T) = \{\mathbf{0}\}$, that is $x = y$. Hence T is one-to-one. □

Remark 2.2 (Geometric picture: the kernel is the direction crushed to zero, the range is the part filled up; Friedberg Figure 2.1(c), p. 66)**.** Take the projection $P(a_1, a_2) = (a_1, 0)$. Every point on the vertical direction is crushed onto the origin, so $\mathcal{N}(P) = \{(0, a_2)\}$ is the y-axis; every point of the x-axis is the image of something, so $\mathcal{R}(P) = \{(a_1, 0)\}$ is the x-axis. The kernel is the direction that gets discarded, the range is what is left of the plane.



The projection $P(a_1, a_2) = (a_1, 0)$: the blue band is the range $\mathcal{R}(P)$ (the x-axis), the red band is the kernel $\mathcal{N}(P)$ (the y-axis); $\boldsymbol{u}$ is pushed straight down onto the x-axis, and $\boldsymbol{u} - P(\boldsymbol{u})$ lies in the kernel.

3 The Matrix Representation of a Linear Transformation

Definition 3.1 (Ordered basis; Friedberg, §2.2, p. 79)**.** Let V be finite-dimensional. A basis of V **endowed with an order** is called an **ordered basis** for V: a finite sequence of linearly independent vectors of V that generates V.

Remark 3.1 (Why the order cannot be dropped; Friedberg Example 1, p. 79)**.** In $\mathbb{F}^3$, $\{e_1, e_2, e_3\}$ can be regarded as an ordered basis and so can $\{e_2, e_1, e_3\}$, but as ordered bases $\beta \neq \gamma$, their first vectors differ. The definition of $[T]_\beta^\gamma$ below says that "column j corresponds to the jth basis vector", so the order is part of the definition, not a notational convenience.

Definition 3.2 (Coordinate vector; Friedberg, §2.2, p. 80; Lay §4.4, p. 216)**.** Let $\beta = \{u_1, \ldots, u_n\}$ be an ordered basis for a finite-dimensional vector space V. For $x \in V$, the scalars $a_1, \ldots, a_n$ with $x = \sum_{i=1}^n a_i u_i$ are called the **coordinates** of x relative to β, and

$$[x]_\beta = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} \in \mathbb{F}^n$$

is the **coordinate vector** of x relative to β.

Definition 3.3 (Matrix representation; Friedberg, §2.2, p. 80)**.** Let V, W be finite-dimensional with ordered bases $\beta = \{v_1, \ldots, v_n\}$ and $\gamma = \{w_1, \ldots, w_m\}$, and let $T : V \to W$ be linear. By the unique representation theorem, let $a_{1j}, \ldots, a_{mj}$ be the γ-coordinates of $T(v_j)$; the $m \times n$ matrix $A = (a_{ij})$ is called the **matrix representation** of T in the ordered bases β and γ, and is written

$$A = [T]_\beta^\gamma.$$

If $V = W$ and $\beta = \gamma$, we write simply $A = [T]_\beta$.

Remark 3.2 (The standard matrix: Lay's version; Lay §1.9, p. 71)**.** When $\beta = \gamma$ are the standard bases, so that $V = \mathbb{F}^n$ and $W = \mathbb{F}^m$, the matrix $[T]_\beta^\gamma$ is the **standard matrix** A of T, and $T(x) = Ax$. Lay 4e states this as a theorem: for $T : \mathbb{R}^n \to \mathbb{R}^m$ linear there exists a **unique** $m \times n$ matrix A with $T(x) = Ax$ for every x (§1.9, p. 71).

Approach. To find the standard matrix there is no need to expand the formula for $T(x)$, **column j is $T(e_j)$** (substitute $x = e_j$ and read off the result). That T is uniquely determined by A is precisely Theorem 3.1 below (the corollary of Friedberg Theorem 2.6): two linear transformations that agree on a basis agree everywhere.

Example 3.1 (The algorithm for a matrix representation, and the effect of changing basis; data from Friedberg Example 3, p. 81)**.** Let $T : \mathbb{R}^2 \to \mathbb{R}^3$ be linear, $T(a_1, a_2) = (a_1 + 3a_2,\ 0,\ 2a_1 - 4a_2)$, and let $\beta = \{e_1, e_2\}$, $\gamma = \{e_1, e_2, e_3\}$ be the standard ordered bases on the two sides. Find $[T]_\beta^\gamma$, and then find the matrix for $\gamma' = \{e_3, e_2, e_1\}$.

Solution. **(1) Send each basis vector through T.** $T(1, 0) = (1, 0, 2)$ and $T(0, 1) = (3, 0, -4)$; since γ is the standard basis, these two vectors are already their own γ-coordinates.

(2) Assemble the matrix. Column j is $[T(v_j)]_\gamma$, hence

$$[T]_\beta^\gamma = \begin{bmatrix} 1 & 3 \\ 0 & 0 \\ 2 & -4 \end{bmatrix}.$$

(3) Change the codomain basis. In $\gamma' = \{e_3, e_2, e_1\}$ the vector (y_1, y_2, y_3) has coordinates $(y_3, y_2, y_1)^\top$. Thus $[T(1, 0)]_{\gamma'} = (2, 0, 1)^\top$ and $[T(0, 1)]_{\gamma'} = (-4, 0, 3)^\top$, giving

$$[T]_\beta^{\gamma'} = \begin{bmatrix} 2 & -4 \\ 0 & 0 \\ 1 & 3 \end{bmatrix}.$$

Not one letter of T has changed; only the way coordinates are read has, and the columns come out reordered. □

Approach. To build a matrix representation, send each vector of the *domain* basis through T and expand the result in the *codomain* basis; the coordinates of the result are one column of the matrix. Changing the codomain basis leaves T itself untouched, so every column has to be written again in the new coordinates.

Theorem 3.1 (A transformation is uniquely determined by its values on a basis; Friedberg Theorem 2.6, §2.1, p. 72)**.** *Let V, W be vector spaces over $\mathbb{F}$ and let $\{v_1, \dots, v_n\}$ be a basis for V. Given $w_1, \dots, w_n \in W$, there exists **exactly one** linear transformation $T : V \to W$ with $T(v_j) = w_j$ for $j = 1, \dots, n$.*

Proof. **Existence:** for $x \in V$, expand $x = \sum_{i=1}^{n} a_i v_i$ in the basis and define $T(x) = \sum_{i=1}^{n} a_i w_i$.
(i) T is linear: write $u = \sum b_i v_i$ and $v = \sum c_i v_i$; then $du + v = \sum (db_i + c_i) v_i$, so

$$T(du + v) = \sum_{i=1}^{n} (db_i + c_i) w_i = d \sum_{i=1}^{n} b_i w_i + \sum_{i=1}^{n} c_i w_i = dT(u) + T(v).$$

(ii) $T(v_j) = w_j$: in the expansion all coefficients except the jth are zero.
Uniqueness: let $U : V \to W$ be linear with $U(v_j) = w_j$. For $x = \sum a_i v_i$, $U(x) = \sum a_i U(v_i) = \sum a_i w_i = T(x)$, so $U = T$. □

Definition 3.4 (The space of linear transformations; Friedberg, §2.2, p. 82)**.** Let V, W be vector spaces over $\mathbb{F}$. The set of all linear transformations from V to W is written $\mathcal{L}(V, W)$, and $\mathcal{L}(V)$ when $V = W$. Addition and scalar multiplication are defined pointwise: $(T+U)(x) = T(x)+U(x)$ and $(aT)(x) = aT(x)$.

Proposition 3.2 (Sum and scalar multiples of transformations; Friedberg Theorem 2.7, §2.2, p. 82)**.** *Let V, W be vector spaces over $\mathbb{F}$, let $T, U : V \to W$ be **linear**, and let $a \in \mathbb{F}$. Then*

(a) for every $a \in \mathbb{F}$, $aT + U$ is a linear transformation;
(b) $\mathcal{L}(V, W)$ is a vector space over $\mathbb{F}$ under these operations.

Proof. (a) Let $x, y \in V$ and $c \in \mathbb{F}$. Then

$$\begin{aligned}(aT + U)(cx + y) &= aT(cx + y) + U(cx + y) \\ &= a[cT(x) + T(y)] + cU(x) + U(y) = c(aT + U)(x) + (aT + U)(y).\end{aligned}$$

The first equality of each line uses the linearity of T and U, and the last one is just the distributive law. By proposition 1.1(2), $aT + U$ is linear.

(b) The zero transformation $T_{\mathbf{0}}$ plays the role of the zero vector, and the remaining eight axioms are checked exactly as for a vector space; that $T + U$ is linear is (a) with $a = 1$. □

Theorem 3.3 (The matrix representation preserves addition and scalar multiplication; Friedberg Theorem 2.8, §2.2, p. 82–83)**.** *Let V, W be finite-dimensional with ordered bases β, γ, and let $T, U : V \to W$ be linear. Then for every $a \in \mathbb{F}$,*

$$[T + U]_\beta^\gamma = [T]_\beta^\gamma + [U]_\beta^\gamma, \qquad [aT]_\beta^\gamma = a[T]_\beta^\gamma.$$

Proof. Let $\beta = \{v_1, \dots, v_n\}$ and $\gamma = \{w_1, \dots, w_m\}$. There exist unique a_{ij}, b_{ij} with $T(v_j) = \sum_{i=1}^{m} a_{ij} w_i$ and $U(v_j) = \sum_{i=1}^{m} b_{ij} w_i$. Hence

$$(T + U)(v_j) = \sum_{i=1}^{m} (a_{ij} + b_{ij}) w_i,$$

so $\left([T + U]_\beta^\gamma\right)_{ij} = a_{ij} + b_{ij} = \left([T]_\beta^\gamma + [U]_\beta^\gamma\right)_{ij}$. The proof of (b) is the same. □

4 A Worked Example

Example 4.1 (Kernel, range, nullity, rank, injectivity and surjectivity: a worked application; Friedberg, Theorem 2.2 and the dimension theorem, §2.1, p. 68–70; Lay's standard matrix, §1.9, p. 71)**.** Let $T : \mathbb{R}^3 \to \mathbb{R}^2$ be linear, $T(x_1, x_2, x_3) = (x_1 - x_2 + x_3,\ x_1 + x_2 - x_3)$. Find bases of $\mathcal{N}(T)$ and $\mathcal{R}(T)$ together with their dimensions, decide whether T is one-to-one and whether it is onto, and find the matrix $[T]_\beta^\gamma$ for the ordered bases $\beta = \{(1,1,0),(1,0,1),(0,1,1)\}$ and $\gamma = \{(1,1),(1,-1)\}$.

Solution. **(1) Verify linearity.** Let $\boldsymbol{x} = (b_1, b_2, b_3)$, $\boldsymbol{y} = (d_1, d_2, d_3)$ and $c \in \mathbb{R}$. Then

$$\begin{aligned} T(c\boldsymbol{x} + \boldsymbol{y}) &= \big(cb_1 + d_1 - cb_2 - d_2 + cb_3 + d_3,\ cb_1 + d_1 + cb_2 + d_2 - cb_3 - d_3\big) \\ &= c(b_1 - b_2 + b_3,\ b_1 + b_2 - b_3) + (d_1 - d_2 + d_3,\ d_1 + d_2 - d_3) \\ &= cT(\boldsymbol{x}) + T(\boldsymbol{y}), \end{aligned}$$

so T is linear by proposition 1.1(2).

(2) Standard matrix. Column j is $T(e_j)$: $T(1,0,0) = (1,1)$, $T(0,1,0) = (-1,1)$, $T(0,0,1) = (1,-1)$, so

$$A = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 1 & -1 \end{bmatrix}.$$

(3) Kernel. Row reduce A:

$$\left[\begin{array}{cc|c} 1 & -1 & 1 \\ 1 & 1 & -1 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & -1 & 1 \\ 0 & 2 & -2 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & -1 \end{array}\right].$$

Thus $x_1 = 0$ and $x_2 = x_3$, with x_3 free:

$$\mathcal{N}(T) = \{(0,t,t) : t \in \mathbb{R}\}, \qquad \text{basis} = \{(0,1,1)^\top\}, \qquad \text{nullity}(T) = 1.$$

(4) Range. $\mathcal{R}(T) = \operatorname{Col} A = \operatorname{Span}\{(1,1)^\top, (-1,1)^\top\}$: the third column $(1,-1)^\top$ is -1 times the second, and the first two are not proportional, so this is the whole span, and

$$\text{basis} = \{(1,1)^\top, (-1,1)^\top\}, \qquad \operatorname{rank}(T) = 2.$$

(5) One-to-one and onto. By theorem 2.5, $\mathcal{N}(T) \neq \{\mathbf{0}\}$, so T is **not one-to-one**; and $\mathcal{R}(T) = \mathbb{R}^2 = W$, so T **is onto**. Here $\dim \mathbb{R}^3 = 3 \neq 2 = \dim \mathbb{R}^2$, so "onto but not one-to-one" does not conflict with theorem 2.5: that theorem equates injectivity with a trivial kernel, while surjectivity asks separately whether $\mathcal{R}(T)$ fills the codomain.

(6) The matrix for β, γ. Send each vector of β through T and expand the result in γ:

$$\begin{aligned} T(1,1,0) &= (0,2) = 1 \cdot (1,1) - 1 \cdot (1,-1), \\ T(1,0,1) &= (2,0) = 1 \cdot (1,1) + 1 \cdot (1,-1), \\ T(0,1,1) &= (0,0). \end{aligned}$$

The three columns are therefore $(1,-1)^\top$, $(1,1)^\top$, $(0,0)^\top$, so

$$[T]_\beta^\gamma = \begin{bmatrix} 1 & 1 & 0 \\ -1 & 1 & 0 \end{bmatrix}.$$

The third column is a zero column because $(0,1,1)^\top \in \mathcal{N}(T)$, a zero column is the matrix's way of saying that this basis vector lies in the kernel. □

Approach. Given $T : \mathbb{R}^n \to \mathbb{R}^m$ by a coordinate formula, **write the standard matrix A first (column j is $T(e_j)$)**, then solve $A\boldsymbol{x} = \mathbf{0}$ once to get the kernel, that is the row reduction of Lecture 1. The range is $\operatorname{Col} A$, and taking the **pivot columns** of the echelon form gives a basis for it. The dimension theorem is the check, $\mathcal{N}(T) = \{\mathbf{0}\}$ decides injectivity, and $\mathcal{R}(T) = \mathbb{R}^m$ decides surjectivity.

5 Supplement: notions Homework 3 uses but the lecture did not cover

Remark 5.1 (Where this comes from)**. T-invariant subspaces** and the **restriction operator** T_W are not part of the lecture's main thread, yet Homework 3 uses them (Problems 14 and 15): Definition 1 of the sheet; Friedberg, §5.4 (printed p. 313).

Definition 5.1 (T-invariant subspace and restriction; Definition 1 of the sheet, Friedberg, §5.4, p. 313)**.** Let W be a subspace of V and $T : V \to V$ linear. If $T(x) \in W$ for every $x \in W$, that is $T(W) \subseteq W$, then W is **T-invariant**; the map $T_W : W \to W$ defined by $T_W(x) = T(x)$ is then the **restriction** of T to W.

6 Checklist and Exercises

By the end of this lecture you should be able to:

1. write down the two axioms of a linear transformation and combine them into the single testing equation $T(cx + y) = cT(x) + T(y)$;
2. recognise the signals, a constant term, $|x_i|$, $x_i x_j$, and say which of (a) or (b) they break;
3. explain why the null space and the range are subspaces, and why the range half of the proof has to pull representatives back first;
4. identify $\mathcal{N}(T)$ with $\operatorname{Nul} A$ and $\mathcal{R}(T)$ with $\operatorname{Col} A$ for $T(x) = Ax$, and compute a basis for each by row reduction, pivot columns included;
5. use "$\mathcal{N}(T) = \{\mathbf{0}\}$" to decide injectivity, and say why surjectivity instead asks whether $\mathcal{R}(T)$ already fills the codomain (equivalently, whether the images of a basis of V span W);
6. state the dimension theorem and say why the proof needs the step "extend a basis of $\mathcal{N}(T)$ to a basis of V";
7. write $[T]^{\gamma}_{\beta}$ on your own, column j is $[T(v_j)]_{\gamma}$, and say which part of the matrix changes when the codomain basis changes;
8. use theorem 3.1 to justify the sentence "T is uniquely determined by A";
9. read a zero column and a non-zero column of a matrix representation: one basis vector lies in the kernel, the other does not.

Exercises (printed pages). Lay: §1.8 p. 68 (Exercises 9–12 on null space and range, 21–22 true/false, 23–33 on linearity), §1.9 p. 78, §4.2 p. 205, §4.4 p. 222, §5.4 p. 293; Friedberg: §2.1 p. 74 (Exercises 2–6 computing everything, 9 non-linear counterexamples, 15–18 one-to-one and onto, 20–21 shift operators, 24–27 projections), §2.2 p. 84 (Exercises 2–5 computing matrix representations, 10–11). Each section also has Practice Problems; do those first.

How to use the two texts together:

- **Geometry and intuition from Lay, theory from Friedberg.** Lay's projection, reflection and shear give the picture of what a linear transformation looks like and supply the standard matrix as a computable entry point; the theory of the null space and the range, nullity, rank, the dimension theorem, the injectivity test, exists only in Friedberg.
- The chain of this lecture: *preserving addition and scalar multiplication → null space and*

range are subspaces → the images of a basis span the range → the dimension theorem balances the two dimensions → an ordered basis turns T into a matrix.

- Theorem 3.1 (Friedberg's Theorem 2.6) is the one result of this lecture worth memorising: it replaces "linearity" by "values on a basis", and every question about T then becomes a question about the columns of a matrix.